Mobility-Aware UAV Positioning and Secret Key Rate Optimization for HAP–UAV CV-QKD

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***Abstract*—This paper investigates mobility-aware unmanned aerial vehicle (UAV) positioning in a high-altitude-platform (HAP)-to-UAV free-space optical continuous-variable quantum key distribution (CV-QKD) system under cloud blockage. Probabilistic constellation shaping is optimized over a finite source catalog to maximize the clear-sky secret key rate (SKR) at each candidate UAV position. We first identify the best stationary position and then formulate a mobility-aware trajectory optimization problem that balances the average usable SKR against a normalized flight-distance penalty subject to a per-slot speed constraint. The resulting finite-horizon discrete problem is solved exactly by dynamic programming on the candidate-position grid. The evaluation compares source shaping, stationary positioning, and cloud-aware UAV mobility, using greedy positioning as a low-complexity trajectory baseline.**

***Index Terms*—Continuous-variable quantum key distribution (CV–QKD), unmanned aerial vehicle (UAV), free-space optics (FSO), high-altitude platform (HAP), probabilistic constellation shaping (PCS), dynamic programming (DP).**

I. HAP–UAV CV-QKD System and Problem Formulation

A. Mobility-Aware UAV Positioning and SKR Optimization

We consider one UAV at a fixed altitude over a $1\,\mathrm{km}\times 1\,\mathrm{km}$ area, discretized into $J = 100$ candidate positions $\mathcal{Q} = \{\mathbf{q}_j\}_{j=1}^{J}$. The mission has N_t slots of duration Δt. Let $c_{j,n} \in \{0,1\}$ indicate whether the link at position j is blocked by clouds in slot n (1: blocked), where $\mathbf{C} = [c_{j,n}]$ follows spatially and temporally correlated Bernoulli states. The full cloud-state matrix is assumed known over the optimization horizon; therefore, the trajectory solution is an offline benchmark.

1) Source and Stationary-Position Optimization: For an unblocked link, the asymptotic SKR is

$$K_j(\nu, V_A) = \beta_{\mathrm{rec}} I_{AB,j}(\nu, V_A) - \chi_{BE,j}(\nu, V_A). \tag{1}$$

Using a finite source catalog $\mathcal{G}$, the best source at position j is obtained from

$$\mathcal{P}_1: \quad K_j^{\star} = \Big[\max_{(\nu,V_A)\in\mathcal{G}} K_j(\nu, V_A)\Big]^{+}, \tag{2}$$

where $[x]^{+} = \max\{x, 0\}$. Problem $\mathcal{P}_1$ is solved once per position under stationary clear-sky statistics. This model assumes that the transmitter can select the pre-optimized source associated with the UAV's current position. The usable SKR is $S_{j,n} = (1 - c_{j,n})K_j^{\star}$, and the best stationary position is

$$\mathcal{P}_2: \quad j_{\mathrm{fix}}^{\star} = \arg\max_{j} \frac{1}{N_t}\sum_{n=1}^{N_t} S_{j,n}. \tag{3}$$

2) Mobility-Aware Trajectory Optimization: Let j_n be the UAV position in slot n and $d_{i,j} = \|\mathbf{q}_i - \mathbf{q}_j\|_2$. We solve

$$\mathcal{P}_3: \max_{j_1,\ldots,j_{N_t}} \quad \frac{1}{N_t}\sum_{n=1}^{N_t} S_{j_n,n} - \frac{\lambda}{D_{\mathrm{ref}}}\sum_{n=2}^{N_t} d_{j_{n-1},j_n} \tag{4}$$
$$\text{s.t.} \quad d_{j_{n-1},j_n} \le v_{\max}\Delta t, \quad n \ge 2.$$

Here $D_{\mathrm{ref}} > 0$ normalizes distance and $\lambda \ge 0$ (bits/use) sets the mobility penalty. The initial position is free, and flight distance is only a surrogate for mobility cost, not a calibrated propulsion-energy model.

II. Optimization Method

A. Finite-Catalog Search and Stationary Benchmark

Because $\mathcal{G}$ and $\mathcal{Q}$ are finite, $\mathcal{P}_1$ is solved by evaluating the SKR of each catalog source at each position and selecting its maximum. The selected source parameters are

$$(\nu_j^{\star}, V_{A,j}^{\star}) \in \arg\max_{(\nu,V_A)\in\mathcal{G}} K_j(\nu, V_A). \tag{5}$$

This search is globally optimal over $\mathcal{G}$, without claiming optimality over continuous source parameters. Once $K_j^{\star}$ has been obtained, $\mathcal{P}_2$ is solved by evaluating the average usable SKR at all J candidate positions.

B. Greedy Trajectory Baseline

For comparison, we adopt a feasible one-step greedy policy that uses only the cloud states in the current slot. Since the initial position is free, it is initialized by

$$j_1^{\mathrm{G}} \in \arg\max_{j=1,\ldots,J} S_{j,1}. \tag{6}$$

For each subsequent slot $n = 2, \ldots, N_t$, the UAV selects

$$j_n^{\mathrm{G}} \in \arg\max_{j:\, d_{j_{n-1}^{\mathrm{G}},j} \le v_{\max}\Delta t} \left\{ S_{j,n} - \frac{N_t\lambda}{D_{\mathrm{ref}}} d_{j_{n-1}^{\mathrm{G}},j} \right\}. \tag{7}$$

The factor N_t ensures consistency with the objective in (4). Unlike dynamic programming, the greedy policy does not account for future cloud states or subsequent movement opportunities and therefore is not guaranteed to be optimal. Its worst-case complexity is $O(N_t J)$.

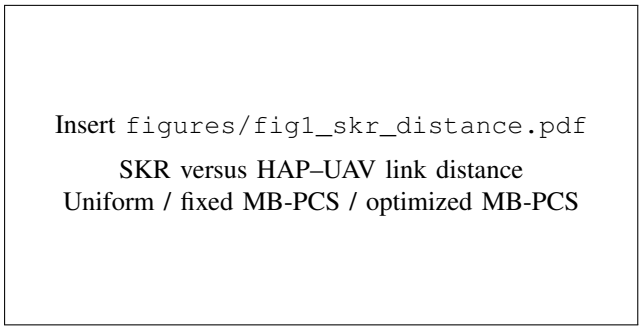


Fig. 1. Secret key rate versus HAP–UAV link distance for uniform, fixed MB-PCS, and optimized MB-PCS source configurations.

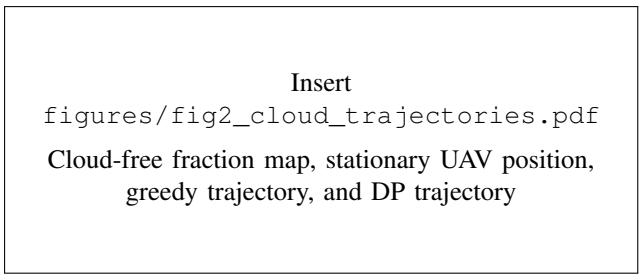


Fig. 2. Cloud-aware UAV positioning over the operating area, including the best stationary position and feasible greedy and DP-based trajectories.

C. Dynamic Programming for UAV Trajectory Optimization

Define the set of feasible predecessor positions for location j as

$$\mathcal{N}(j) = \{i \in \{1, \dots, J\} : d_{i,j} \le v_{\max}\Delta t\}. \tag{8}$$

Let $F_n(j)$ be the maximum cumulative reward over the first n slots among all feasible trajectories ending at position j. Multiplying the objective in (4) by N_t gives the initialization

$$F_1(j) = S_{j,1}, \qquad j = 1, \dots, J, \tag{9}$$

and the recursion, for $n = 2, \dots, N_t$,

$$F_n(j) = S_{j,n} + \max_{i \in \mathcal{N}(j)} \left\{ F_{n-1}(i) - \frac{N_t \lambda}{D_{\text{ref}}} d_{i,j} \right\}. \tag{10}$$

The maximizer at each step is stored as

$$B_n(j) \in \arg\max_{i \in \mathcal{N}(j)} \left\{ F_{n-1}(i) - \frac{N_t \lambda}{D_{\text{ref}}} d_{i,j} \right\}. \tag{11}$$

The optimal objective value is

$$U^\star = \frac{1}{N_t} \max_{j=1,\dots,J} F_{N_t}(j). \tag{12}$$

Starting from $j^\star_{N_t} \in \arg\max_j F_{N_t}(j)$, we recover the optimal trajectory by backward substitution $j^\star_{n-1} = B_n(j^\star_n)$. The Bellman recursion considers every feasible predecessor, so it yields a globally optimal solution to $\mathcal{P}_3$ on the discretized grid for the given cloud states. The worst-case running time is $O(N_t J^2)$, and storing all backpointers requires $O(N_t J)$ memory.

Algorithm 1 Source Search, Greedy Baseline, and DP-Based UAV Positioning

Input: $\mathcal{Q}$, $\mathcal{G}$, $\mathbf{C}$, N_t, Δt, $v_{\max}$, D_{ref}, λ
Output: $\{K_j^\star\}_{j=1}^J$, $j^\star_{\text{fix}}$, $\{j_n^{\text{G}}\}_{n=1}^{N_t}$, $\{j_n^\star\}_{n=1}^{N_t}$

for $j = 1, \dots, J$ **do**
  Evaluate $K_j(\nu, V_A)$ for all $(\nu, V_A) \in \mathcal{G}$
  $K_j^\star \leftarrow [\max_{(\nu,V_A)\in\mathcal{G}} K_j(\nu, V_A)]^+$
  for $n = 1, \dots, N_t$ **do**
    $S_{j,n} \leftarrow (1 - c_{j,n}) K_j^\star$
  end for
end for
$j^\star_{\text{fix}} \leftarrow \arg\max_j \sum_{n=1}^{N_t} S_{j,n}$
$j_1^{\text{G}} \leftarrow \arg\max_j S_{j,1}$
for $n = 2, \dots, N_t$ **do**
  $j_n^{\text{G}} \leftarrow \arg\max_{j:\, d_{j_{n-1}^{\text{G}},j} \le v_{\max}\Delta t} \left\{ S_{j,n} - \frac{N_t\lambda}{D_{\text{ref}}} d_{j_{n-1}^{\text{G}},j} \right\}$
end for
for $j = 1, \dots, J$ **do**
  $F_1(j) \leftarrow S_{j,1}$
end for
for $n = 2, \dots, N_t$ **do**
  for $j = 1, \dots, J$ **do**
    $B_n(j) \leftarrow \arg\max_{i \in \mathcal{N}(j)} \left\{ F_{n-1}(i) - \frac{N_t\lambda}{D_{\text{ref}}} d_{i,j} \right\}$
    $F_n(j) \leftarrow S_{j,n} + F_{n-1}(B_n(j)) - \frac{N_t\lambda}{D_{\text{ref}}} d_{B_n(j),j}$
  end for
end for
$j^\star_{N_t} \leftarrow \arg\max_j F_{N_t}(j)$
for $n = N_t, N_t - 1, \dots, 2$ **do**
  $j^\star_{n-1} \leftarrow B_n(j^\star_n)$
end for

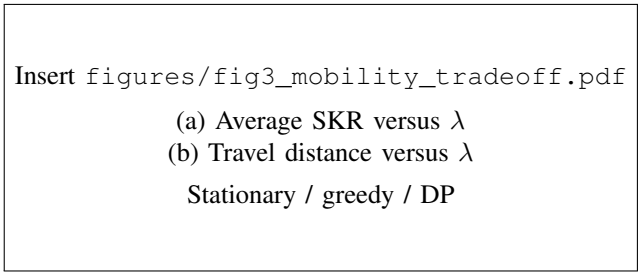


Fig. 3. SKR–mobility tradeoff versus the distance penalty λ: (a) average usable SKR and (b) total UAV travel distance.

III. Numerical Evaluation

We compare (i) a stationary UAV at $j^\star_{\text{fix}}$, (ii) a feasible greedy policy that selects the next position using the current-slot SKR and movement penalty as defined in (7), and (iii) the globally optimal offline DP trajectory. All mobility methods use the same cloud realization, source catalog, and speed

constraint. For a trajectory $\boldsymbol{j} = (j_1, \ldots, j_{N_t})$, we report

$$\overline{K}(\boldsymbol{j}) = \frac{1}{N_t} \sum_{n=1}^{N_t} S_{j_n,n}, \tag{13}$$

$$D(\boldsymbol{j}) = \sum_{n=2}^{N_t} d_{j_{n-1},j_n}, \tag{14}$$

$$U(\boldsymbol{j}) = \overline{K}(\boldsymbol{j}) - \frac{\lambda}{D_{\mathrm{ref}}} D(\boldsymbol{j}). \tag{15}$$

A. Impact of Probabilistic Constellation Shaping

Fig. 1 is intended to compare optimized uniform 256-QAM, fixed MB-shaped 256-QAM, and optimized MB-shaped 256-QAM under the same physical channel assumptions. The operational transmission range can be identified from the positive-SKR region.

B. UAV Trajectories under Correlated Cloud Blockage

Fig. 2 is intended to show the spatial cloud-free fraction $N_t^{-1} \sum_{n=1}^{N_t} (1 - c_{j,n})$, the best stationary position, and the greedy and DP trajectories over the $1\,\mathrm{km} \times 1\,\mathrm{km}$ operating region.

C. SKR–Mobility Tradeoff

Fig. 3 is intended to quantify the impact of the mobility penalty λ on average usable SKR and flight distance. The stationary, greedy, and DP methods should be evaluated using the same cloud states and physical parameters.

IV. Conclusion

This work formulates a finite-catalog source optimization and mobility-aware positioning framework for cloud-affected HAP–UAV CV-QKD. Source parameters and stationary positioning are selected by finite search, while the discrete trajectory is optimized exactly by dynamic programming for a known cloud-state horizon. The proposed evaluation separates shaping gain from mobility gain and explicitly accounts for the tradeoff between usable SKR and UAV travel distance. Online decision-making with uncertain future cloud states remains an extension beyond the present model.